JEFFREY HAMLIN VINOD AUXALIA RAJA
(+353) 0894409273 | jeffreyvhamlin@gmail.com | LinkedIn: https://www.linkedin.com/in/jeffrey-hamlin-v
Location: Dublin, Ireland (Hybrid | Eligible to work in Ireland)

PROFESSIONAL SUMMARY

I am an MSc Computer Science (Data Science) student at Trinity College Dublin with hands-on experience in data analytics, automation, machine learning and applied AI supported by solid foundation in software development. I am a curious and quick learner who thrives in collaborative environments, adapting easily to new challenges through clear communication and teamwork. I have built practical, data-driven solutions using Python, Power BI and UiPath Studio, applying analytical thinking and structured problem-solving to real-world use cases. I also value giving back to the community and bring a collaborative, reliable and people-oriented approach shaped through volunteering and leadership experiences.

SKILLS

- **Data Analytics & Visualization:** Python, SQL, Power BI, D3, Microsoft Excel, Data Cleaning, Exploratory Data Analysis, Dashboard Design, Data-Driven Reporting
- **Machine Learning & AI(Applied):** Supervised & Unsupervised Learning, Ensemble Models, Computer Vision (YOLOv8, CNNs), LSTMs, Sentiment Analysis, Model Evaluation, Feature Engineering
- **Programming & Software Development**: Python, R, Java, C, HTML, CSS, Bootstrap, REST APIs, Git
- **Automation & Workflow Tools**: UiPath Studio, Power Automate, Process Automation, Queue-Based Workflows
- **Data Management**: MySQL, SQL Server, Relational Databases, Data Modelling, Microsoft Excel, Google Sheets
- **Tools & Collaboration**: GitHub, Agile Methodologies, Team Collaboration, Technical Documentation, G-Suite Tools

PROJECTS

Real Time Traffic Prediction & Signaling System (Undergrad - Final Year Project)

- Developed an AI-powered system optimized for commercial scalability using YOLOv8 and LSTM, achieving >90% detection accuracy and predictive congestion alerts.
- Deployed via Django for real-time adaptive signaling, reducing traffic waiting time by ~25% using SUMO simulations.
- Collaborated in a 3-member agile team delivering a scalable, sustainable smart-city solution integrating analytics and automation.

Scalable AI Captcha Systems

Phase 1: AI CAPTCHA Recognition on Edge Devices

- Designed and trained a CRNN-based CAPTCHA recognition model using a custom dataset of 4,000+ synthetic CAPTCHA images with varied fonts, symbols and noise.
- Performed character set analysis and dataset curation, followed by model optimization and conversion to TensorFlow Lite for deployment on a compute and memory-constrained Raspberry Pi as part of a scalability challenge.
- Deployed and evaluated the system on physical Raspberry Pi hardware, analyzing inference latency, accuracy and limitations imposed by constrained and faulty devices.

Phase 2: Human-vs-AI CAPTCHA Using Behavioral & Semantic Scoring

- Designed an AI-resistant CAPTCHA system that distinguishes humans from automated agents using behavioral telemetry and semantic analysis, implemented to run on a resource-constrained Raspberry Pi server for scalability.
- Implemented interaction telemetry capture and feature reconstruction (keystroke timing, typing pauses, mouse movement patterns, paste events and tab visibility) under edge-device constraints.
- Evaluated robustness using multiple automated attacker scripts and refined transparent, rule-based heuristics to generate interpretable pass / fail / retry decisions on constrained hardware.

WhatsApp Message Automation

- Created a UiPath bot to send personalized WhatsApp messages to multiple recipients from UiPath Cloud queues.
- Gained skills in UiPath Studio, automation workflows and queue-based bot deployment.
- Improved efficiency as HR of a student club by automating large-scale communication and reducing manual effort by more than 70%.

Sentiment Analysis for College Reviews

- Collected and analyzed placement-related reviews from Shiksha to evaluate student opinions.
- Designed a sentiment classifier and automated personalized replies - thanking positive feedback, addressing negative reviews and affirming improvements for neutral feedback.
- Enhanced engagement by ensuring timely acknowledgment and addressing concerns effectively.

EAFC 26 Footballers Rating Dashboard

- Developed an interactive dashboard using D3 to analyze metric of 17,000+ football players, including ratings, potential and market value.
- Executed data cleaning, transformation and interactive visualization design, applying strong quantitative skills for exploratory analytics.
- Completed the project out of curiosity, to learn and strengthen data visualization skills while gaining hands-on experience in dashboard design and exploratory analytics

EDUCATION

Trinity College Dublin, Ireland - MSc Computer Science - Data Science | Sep 2025 – Aug 2026 (Expected)
Rajalakshmi Engineering College, India- B.E. Computer Science and Engineering | 2021-2025
Key Modules *- Cloud Computing, Data Analytics, Software Engineering, Scalable Computing*

WORKSHOPS & INTERNSHIPS

Deep Learning Concepts Workshop, University of Texas, Dallas **(August 2024)**

- Gained foundational knowledge in deep learning under Dr. Richard K. Min, covering neural networks, backpropagation and model optimization.
- Completed hands-on training using Iris and Fashion MNIST datasets for classification tasks.
- Strengthened understanding of advanced machine learning concepts and practical applications in real-world datasets.

Frontend Developer Intern, Yi Chennai **(August 2023 – Oct 2023)**

- Collaborated in a four-member team, taking primary responsibility for end-to-end frontend development and deployment of the Yi Yuva website.
- Acquired expertise in responsive web design, HTML, CSS, and Bootstrap.
- Delivered a fully responsive website with improved usability for 500+ users, ensuring smooth navigation without the need for backend integration.

LEADERSHIP/VOLUNTEERING

- **Yi Yuva REC -** Progressed from Human Resource team member 2022 to Chair – Sports Vertical 2025, demonstrating leadership, team-management and communication skills while coordinating volunteers and managing logistics for large-scale initiatives. Organized and executed 50+ community-focused events including placement guidance sessions, autism-awareness activities, fundraisers and animal welfare programs while maintaining a strong commitment to diversity and inclusion and ensuring smooth, well-planned execution across teams.
- **Recharge (National Intercollegiate Cultural Fest) -** Progressed from Content Team (2023) to Promotions & Sponsor Team (2024) and core leadership role (2025), demonstrating strategic planning, decision-making and organizational abilities in managing large-scale cultural initiatives. Led and coordinated a 150+ member crew, overseeing event promotion across colleges, creating engaging social media content, preparing celebrity bytes and handling sponsor contracts — fostering teamwork, communication and seamless event execution.
- **VDP Trinity College Dublin** - Member of Trinity Vincent de Paul (VDP), Ireland's largest student charity and player for its social football club - Galloping Stallions, demonstrating a balance between academics, community engagement and sporting passion.